ITC4214A1 - Internet Programming

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Coursework - Report

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Project Concept & Purpose

So, the website I have created it is called AeroTask Pro and is a front-end, responsive web application designed as a “Command Center” for commercial drone operators. The platform bridges the gap between field pilots and ground control by providing a centralized dashboard to manage flight missions, monitor fleet telemetry and track hardware maintenance.

Key Mechanisms

- Data Persistence” Tasks are stored as stringified JSON objects in localStorage under the key “aeroTasks”/ This ensures data is retained across page reloads and browser sessions.
- Dual-View Architecture: The system utilizes a unified data source array that dynamically renders into both a tabular list view and a drag-and-drop Kanban boards.
- CRUD Operations: User can create, read, update and delete missions. Each entry contains a unique ID, task name, description, date, priority level, and status

Documentation of Coding Decisions

The Tasks Page (tasks.html & tasks.js)

- To optimize performance, event listeners for dynamically generated buttons (edit, complete and delete) were attached to the parent container. This prevents memory leaks and ensures buttons work immediately upon rendering.
- Drag-and-Drop events (drag start, drag over, drop) were implemented for the Kanban boards. The DataTransfer object passes the specific ID, updating the status in memory upon a successful drop.
- Recognizing that drag-and-drop is difficult on mobile devices, a responsive fallback was implemented. Tapping a card on mobile devices triggers a Bootstrap Modal, allowing users to move tasks between columns via button clicks.

Latest Activity Section (index.html)

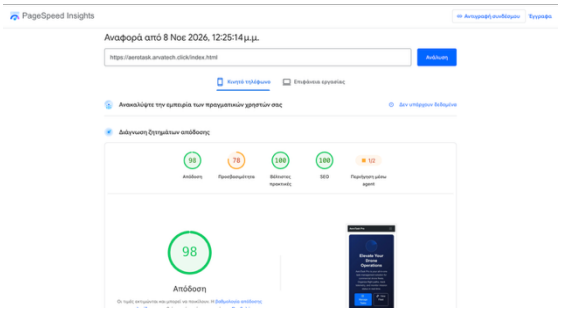
- The homepage pulls the latest pending tasks directly from localStorage, rendering a “Recent/High-Priority Tasks” widget without requiring a backeend database.
- The Open-Meteo API was integrated using AJAX to fetch live weather conditions (temperature , wind speed). A decision engine evaluates this data against flight rules to simulate real-world drone operations constraints.

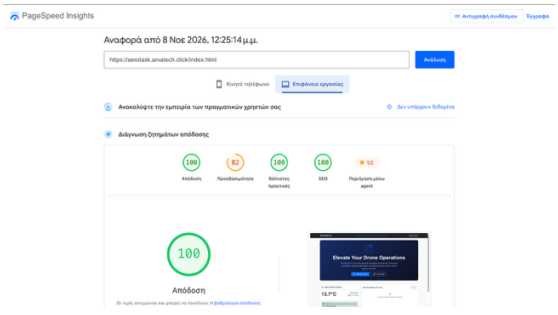
GitHub Repository

- The complete source code and commit history can be found here:

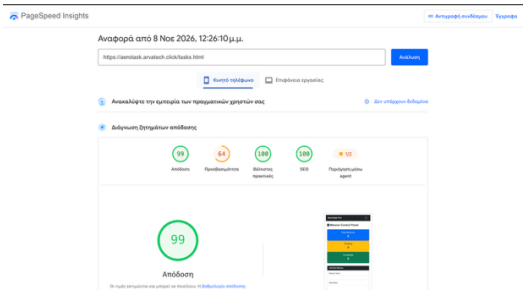
  https://github.com/ArVa34/MidTerm_Exam

Chrome Lighthouse Audit and Reflection

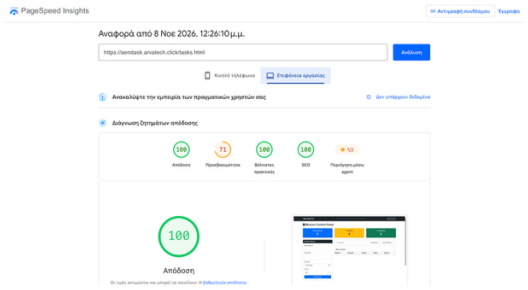
1) 

2) 

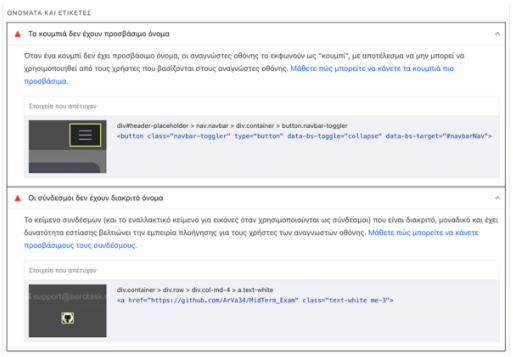
3)


PageSpeed Insights
Αναφορά από 8 Νοε 2026, 12:26:10 μ.μ.
https://aerotask.arvatech.click/tasks.html
Ανάλυση
Κινητό τηλέφωνο
Επιφάνεια εργασίας
99
Απόδοση

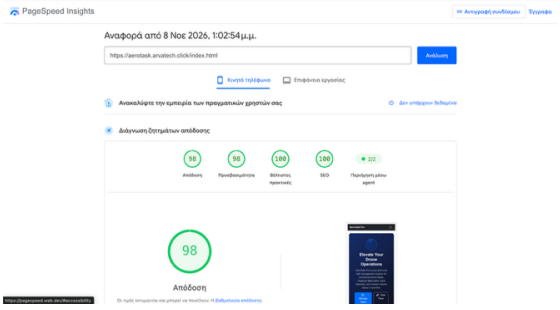
4)


PageSpeed Insights
Αναφορά από 8 Νοε 2026, 12:26:10 μ.μ.
https://aerotask.arvatech.click/tasks.html
Ανάλυση
Επιφάνεια εργασίας
100
Απόδοση

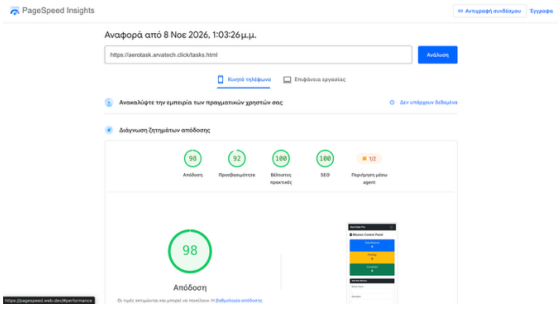
5)


ΟΝΟΜΑΤΑ ΚΑΙ ΕΤΙΚΕΤΕΣ
Τα κουμπιά δεν έχουν προσβάσιμο όνομα
Όταν ένα κουμπί δεν έχει προσβάσιμο όνομα, οι αναγνώστες οθόνης το εκφωνούν ως "κουμπί", με αποτέλεσμα να μην μπορεί να χρησιμοποιηθεί από τους χρήστες που βασίζονται στους αναγνώστες οθόνης. Μάθετε πώς μπορείτε να κάνετε τα κουμπιά πιο προσβάσιμα.
Στοιχεία που απέτυχαν
div#header-placeholder > nav.navbar > div.container > button.navbar-toggler
<button class="navbar-toggler" type="button" data-bs-toggle="collapse" data-bs-target="#navbarNav">
Οι σύνδεσμοι δεν έχουν διακριτό όνομα
Το κείμενο συνδέσμων (και το εναλλακτικό κείμενο για εικόνες όταν χρησιμοποιούνται ως σύνδεσμοι) που είναι διακριτό, μοναδικό και έχει δυνατότητα εστίασης βελτιώνει την εμπειρία πλοήγησης για τους χρήστες των αναγνωστών οθόνης. Μάθετε πώς μπορείτε να κάνετε προσβάσιμους τους συνδέσμους.
Στοιχεία που απέτυχαν
div.container > div.row > div.col-md-4 > a.text-white
<a href="https://github.com/ArYa34/MidTerm_Exam" class="text-white me-3">
support@aerotask

6)


PageSpeed Insights
Αναφορά από 8 Νοε 2026, 1:02:54 μ.μ.
https://aerotask.arvatech.click/index.html
Ανάλυση
Κινητό τηλέφωνο
Επιφάνεια εργασίας
98
Απόδοση

7)


PageSpeed Insights
Αναφορά από 8 Νοε 2026, 1:03:26 μ.μ.
https://aerotask.arvatech.click/tasks.html
Ανάλυση
Κινητό τηλέφωνο
Επιφάνεια εργασίας
98
Απόδοση

- The site performs well due to the use of Vanilla JS and jQuery, as well as serving assets like Bootstrap via high-speed CDNs.
- Semantic HTML5 elements and proper contrast ratios in Dark Mode contributed to high accessibility scores. ARIA labels were added to the mobile hamburger and close buttons
- Extensive open graph and twitter car metadata was injected into that head tags across all pages. Comprehensive meta descriptions and structures title tags enhance search engine crawlability.

Accessibility Enhancements

- Problem: The lighthouse audit flagged that several interactive elements lacked accessible names making them invincible to screen readers. This included the mobile hamburger menu button and the GitHub social link.
- Solution Added descriptive aria-label attributes to the elements. That instantly resolved the screen reader warnings.

Reflections on Development Challenges

Challenge 1: Mobile Layout Overflow

Problem: On the contact page, the Google Maps iframe caused horizontal scrollbar on mobile devices, breaking the responsive layout.

Solution. Identified that the Bootstrap “g-5” class was pushing negative margins off-screen. Replaced it with responsive gutter utility classes to separate vertical and horizontal spacing, resolving the overflow while maintaining desktop proportions.

Challenge 2: Modal UI Freezes with API logic

Problem: When successfully submitting the contact form via EmailJS API, triggering the success modal would sometimes freeze the screen.

Solution: Refactored the DOM manipulation to use “bootstrap.Modal.getOrCreateInstance()” rather than manipulating classes directly via jQuery. This ensured the Bootstrap js engine safely managed the backdrop lifecycle.

Challenge 3: Maintaining UI State Across Pages

Problem: The dark mode toggle and navbar active link highlight would reset to defaults upon navigating to a new page.

Solution: Implemented a robust localStorage strategy to save the dark mode preference (“enable”/”diabasled”). Wrote a dynamic highlightActivePage() function that parses window.locatons.pathname to apply the active class to the correct navigation item dynamically upon page load.

Challenge 4: Layout Shifts

Problem: Navigating through the pages caused some white flashes to occur before the dark mode was loaded, because the external deferred JavaScript after the initial paint. Furthermore, loading the header via jQuery caused a layout shift where the main content jumped down once the navbar was rendered.

Solution: The white flash was eliminated by placing an asynchronous inline script immediately after the opening <body> tag to check localStorage and apply the dark mode class before the browser painted the first frame. The layout shift was resolved by assigning a min-height (56px) to the header placeholder, and a CSS fade-in animation was added to the body to mask the loading sequence, creating a smooth transition.

Personal Reflections

So, I think that I would have made a better website and therefore implemented better technologies across several pages (fleet.html, contact.html and index.html) if I had

better schedule during the development of the website. If I were to take each page separately, index.html has some nice features. I am happy that I thought about implementing this idea where if the weather is bad, and the priority of the task is high, a decision engine should send the appropriate message on whether the task should be done or not. The active missions overview, I think that this is okay, nothing more to add to it. I would really like to have implemented some analytics in this page, that would import data from localStorage and show the most important things. On the page of tasks.html, overall, I did a great job however I did not implement any analytics which was part of the assessment. I did not have the time to do so and even though I thought about implementing it at the last minute, the drawbacks of maybe messing things and functions and having to debug close to the deadline was not worth it. For the about.html I think I had nothing else to add. About the fleet.html – this would have a been a very nice page to implement a lot of new things, however due to bad time decision making I kept it very simple, this page has no logic whatsoever, even though it was a good opportunity to connect the tasks page and its logic with drones and create something unique. I have added some extra things, like the 404.page, included EmailJS which I really like it, gives a vide of professionalism, especially with the auto-reply email. About the commits, I think I overdid a lot of commits, however I would like to say that in the end, I connected Hostinger with GitHub so whenever I pushed a commit, the changes would automatically update to my website. At first, I was doing copy, paste and deploy to every single file. Finally, I had to face a lot of challenges and that really helped me to understand better the core mechanisms. I think that in most challenges I did a good job, however if I had implemented a better strategy on how to do certain things from the start, the website would have been better, and more technologies would have been added to it.